THE PICARD GROUP OF A UNIQUE FACTORIZATION DOMAIN IS TRIVIAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The author writes 'g = v prod_{i in J} p_j^{b_j}' using j both as the summation index and as the symbol introduced by the set J; the expression is reproduced unchanged.

Allowed standard prerequisites (editorial).

- Invertible modules are finite locally free of rank one and their duals are inverse. invoked at proof line 36189. Justifies that L is finite locally free and that varphi(L) is an invertible ideal, which is what makes the localization argument available.
- Uniqueness of prime factorization in a UFD and its preservation under localization. used at proof lines 36201-36218. Is the arithmetic input that forces b_j = c_j in the final comparison.
- The cover criterion for an ideal to be generated by a given element. invoked at proof line 36210. Reduces generating I by h to the case where I_a is principal for a single a.

Source reference. Original lemma 00EO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 542–542. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 36118: sources/stacks-more-algebra/pinned-original-source.tex, source lines 36118–36118. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let R be a UFD. Then* $\mathrm{Pic}(R)$ *is trivial.*

Proof. Let L be an invertible R-module. By Lemma 36118 we see that L is a finite locally free R-module. In particular L is torsion free and finite over R. Pick a

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nonzero element $\varphi \in \mathrm{Hom}_R(L, R)$ of the dual invertible module. Then $I = \varphi(L) \subset R$ is an ideal which is an invertible module. Pick a nonzero $f \in I$ and let

$$f = up_1^{e_1} \ldots p_r^{e_r}$$

be the factorization into prime elements with p_i pairwise distinct. Since L is finite locally free there exist $a_i \in R$, $a_i \notin (p_i)$ such that $I_{a_i} = (g_i)$ for some $g_i \in R_{a_i}$. Then p_i is still a prime element of the UFD R_{a_i} and we can write $g_i = p_i^{c_i} g_i'$ for some $g_i' \in R_{a_i}$ not divisible by p_i. Since $f \in I_{a_i}$ we see that $e_i \geq c_i$. We claim that I is generated by $h = p_1^{c_1} \ldots p_r^{c_r}$ which finishes the proof.

To prove the claim it suffices to show that I_a is generated by h for any $a \in R$ such that I_a is a principal ideal (Algebra, Lemma 00EO). Say $I_a = (g)$. Let $J \subset \{1, \ldots, r\}$ be the set of i such that p_i is a nonunit (and hence a prime element) in R_a. Because $f \in I_a = (g)$ we find the prime factorization $g = v \prod_{i \in J} p_j^{b_j}$ with v a unit and $b_j \leq e_j$. For each $j \in J$ we have $I_{aa_j} = gR_{aa_j} = g_j R_{aa_j}$, in other words g and g_j map to associates in R_{aa_j}. By uniqueness of factorization this implies that $b_j = c_j$ and the proof is complete. $\square$